Automobile Price Prediction Using UCI Imports-85 Dataset

Data Cleaning, Exploratory Analysis, and Multiple Linear Regression Modeling

Presenter

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Dataset Overview

UCI Imports-85 dataset

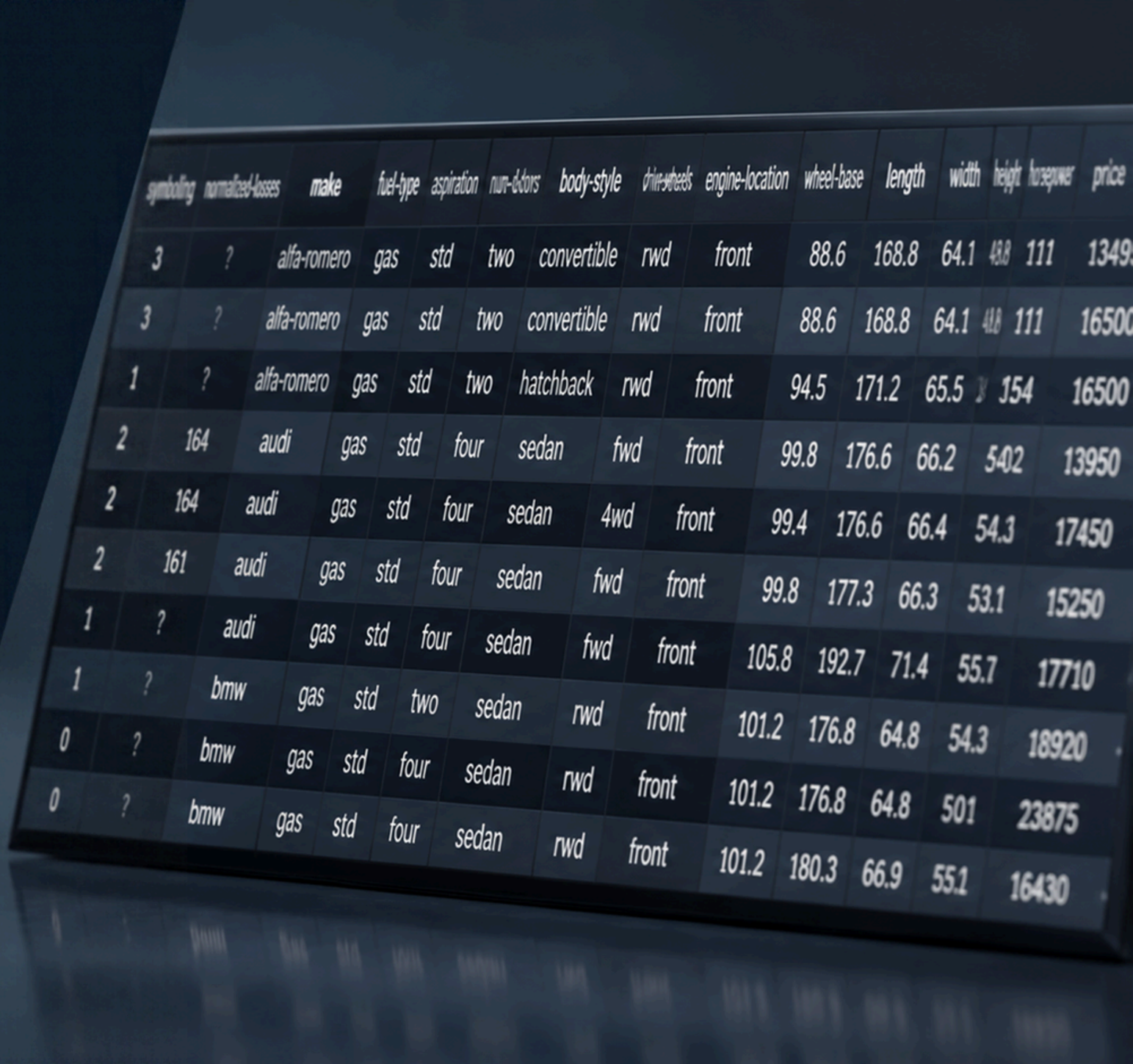
205 vehicles, 26 variables

Target variable
numeric vehicle price in USD

After data cleaning

199 valid observations

Primary predictors
horsepower, width, height



symboling	normalized-losses	make	fuel-type	aspiration	num-doors	body-style	drive-wheels	engine-location	wheel-base	length	width	height	horsepower	price
3	?	alfa-romero	gas	std	two	convertible	rwd	front	88.6	168.8	64.1	48.8	111	1349
3	?	alfa-romero	gas	std	two	convertible	rwd	front	88.6	168.8	64.1	48.8	111	16500
1	?	alfa-romero	gas	std	two	hatchback	rwd	front	94.5	171.2	65.5	[illegible]	154	16500
2	164	audi	gas	std	four	sedan	fwd	front	99.8	176.6	66.2	5402		13950
2	164	audi	gas	std	four	sedan	4wd	front	99.4	176.6	66.4	54.3		17450
2	161	audi	gas	std	four	sedan	fwd	front	99.8	177.3	66.3	53.1		15250
1	?	audi	gas	std	four	sedan	fwd	front	105.8	192.7	71.4	55.7		17710
1	?	bmw	gas	std	two	sedan	rwd	front	101.2	176.8	64.8	54.3		18920
0	?	bmw	gas	std	four	sedan	rwd	front	101.2	176.8	64.8	501		23875
0	?	bmw	gas	std	four	sedan	rwd	front	101.2	180.3	66.9	55.1		16430

Data Cleaning & Preprocessing

01 replaced missing markers with NaN to standardize missing data

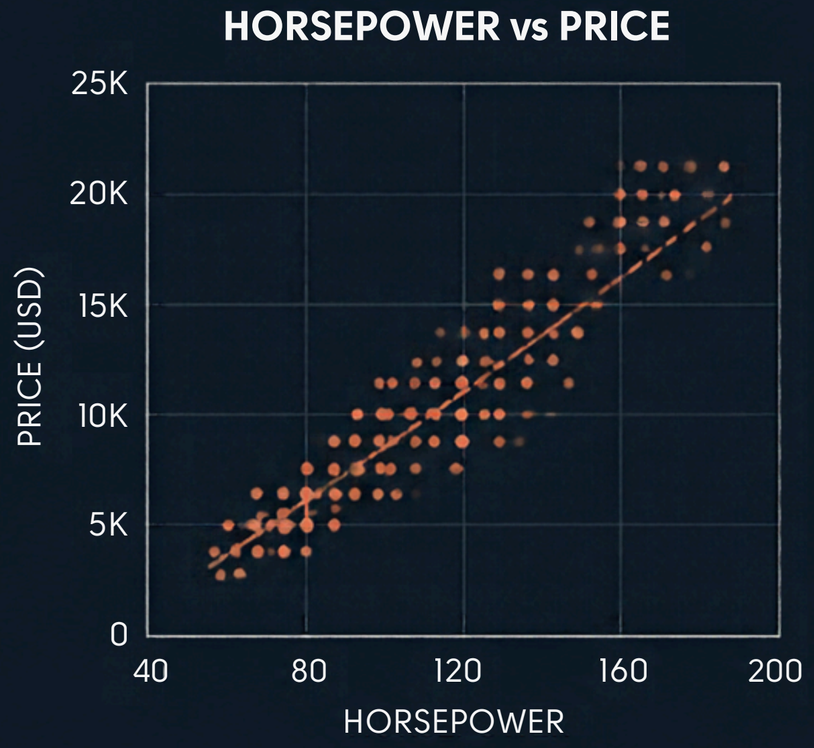
02 Converted categorical fields to numeric types

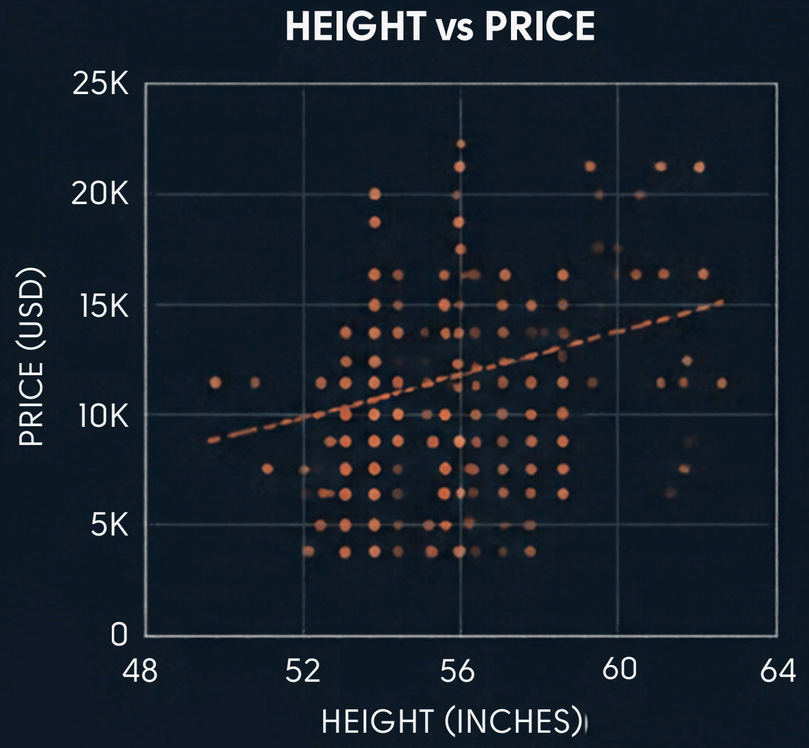
03 Removed rows missing target or key predictors

199 valid observations remain

04 No feature scaling applied to preserve coefficient interpretability

Exploratory Data Analysis





Physical vehicle dimensions and power show **approximate linear relationships** with price.

Modeling Approach & Rationale

MODEL

Multiple Linear Regression
using Ordinary Least Squares

WHY THIS MODEL

Price is a continuous variable

Linear relationships expected between predictors and price

DATA SPLIT

80% training (159 obs)
20% testing (40 obs)
random seed 0

EVALUATION

R^2 score

Model Results & Evaluation

Primary model
horsepower, width, height

Training R^2	Testing R^2
0.748	**0.822**

Key coefficients

Intercept	**-109,617.10**
Horsepower	**+113.74**
Width	**+1,490.94**

Secondary model
symboling, compression-ratio, stroke

Training R^2	Testing R^2
0.004	**0.030**

Poor fit

weak predictors show
poor correlation with price

Power and dimensions **strongly predict price**,
while weak predictors show poor correlation.

Conclusions, Limitations & Future Work

KEY TAKEAWAYS

- Vehicle power and physical size are effective predictors of price
- Model shows robust generalization without overfitting

LIMITATIONS

- No luxury-branding or categorical features included
- Linear model may miss complex feature interactions

FUTURE WORK

- Incorporate brand and luxury metrics
- Explore non-linear models and feature engineering

Thank You
for Your Attention!

Questions and discussion welcome

Reference

Pedregosa et al. (2011),
Scikit-learn: Machine Learning in Python